

LOCALIZING MOTIVES AND SELMER K -THEORY

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ABSTRACT. We construct a quotient of localizing motives by requiring the $\mathbb{A}^1$ -localized unit to be KU -local. Mapping from the unit defines an invariant of small stable categories with a canonical map to Selmer K -theory. We compute the fiber of this map and prove that it is an equivalence on connective ring spectra and on $\mathbb{Q}$ -linear stable categories, as well as after p -completion on $\mathbb{Z}[1/p]$ -linear stable categories. For every prime p , the comparison fails on some small $\mathbb{F}_p$ -linear stable category, which can be realized by a single nonconnective algebra. The obstruction is the noncompactness of a representing motive, together with the continuity of TC/p . At chromatic heights at least two, the comparison is the localized cyclotomic trace; the resulting calculation for $KU_p^\wedge$ distinguishes the two invariants even after p -completion.

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1. INTRODUCTION

Selmer K -theory of a small stable category $\mathcal{C}$ is the pullback

$$(1.1) \quad K^{\mathrm{Sel}}(\mathcal{C}) = L_{KU}K(\mathcal{C}) \times_{L_{KU}\mathrm{TC}(\mathcal{C})} \mathrm{TC}(\mathcal{C}),$$

where K is nonconnective algebraic K -theory and the map $K(\mathcal{C}) \rightarrow \mathrm{TC}(\mathcal{C})$ is the cyclotomic trace [CM, Definition 6.1]. Bhatt–Clausen–Mathew ask for a direct description of this invariant, at least after completion at a prime [BCM, Remark 2.23]. Annala–Iwasa give a geometric description over quasi-compact quasi-separated derived schemes [AI, Theorem 5.4.4]. Here we study a quotient of localizing motives and determine where it recovers (1.1).

The category $\mathrm{Mot}^{\mathrm{loc}}$ of finitary localizing motives has universal invariant $U_{\mathrm{loc}} : \mathrm{Cat}^{\mathrm{perf}} \rightarrow \mathrm{Mot}^{\mathrm{loc}}$ and unit $\mathbf{1} = U_{\mathrm{loc}}(\mathrm{Perf}(\mathbb{S}))$. Polynomial localization $L_{\mathbb{A}^1}^{\mathrm{mot}}$ uses the flat affine line $\mathbb{S}[t]$. It is smashing, so its localized unit $L_{\mathbb{A}^1}^{\mathrm{mot}}\mathbf{1}$ is an idempotent commutative algebra [E–R, Theorem 8.1].

We form the symmetric monoidal quotient that inverts the map

$$(1.2) \quad L_{\mathbb{A}^1}^{\mathrm{mot}}\mathbf{1} \longrightarrow (L_{KU}\mathbb{S}) \otimes L_{\mathbb{A}^1}^{\mathrm{mot}}\mathbf{1}.$$

Thus we kill the localizing tensor ideal generated by its fiber. Writing q for the quotient functor, define

$$(1.3) \quad \begin{aligned} q : \mathrm{Mot}^{\mathrm{loc}} &\longrightarrow \mathrm{Mot}_{\mathrm{Sel}}^{\mathrm{loc}}, \\ K_{\mathrm{mot}}^{\mathrm{Sel}}(\mathcal{C}) &= \mathrm{map}_{\mathrm{Mot}_{\mathrm{Sel}}^{\mathrm{loc}}}(q\mathbf{1}, qU_{\mathrm{loc}}(\mathcal{C})). \end{aligned}$$

The quotient requires the polynomially localized unit to be KU -local. Cyclotomic realization supplies the comparison with Selmer K -theory.

Main Theorem. *There is a natural transformation $\alpha_{\mathcal{C}} : K_{\mathrm{mot}}^{\mathrm{Sel}}(\mathcal{C}) \rightarrow K^{\mathrm{Sel}}(\mathcal{C})$ with the following properties.*

- (i) *If R is a connective E_1 -ring spectrum, then $\alpha_{\mathrm{Perf}(R)}$ is an integral equivalence.*
- (ii) *For every prime p , the map $(\alpha_{\mathcal{C}})_p^{\wedge}$ is an equivalence on all small $\mathbb{Z}[1/p]$ -linear stable categories. On all small $\mathbb{Q}$ -linear stable categories, $\alpha_{\mathcal{C}}$ is an integral equivalence.*
- (iii) *For every prime p , there is a small $\mathbb{F}_p$ -linear stable category $\mathcal{C}$ for which $\alpha_{\mathcal{C}}/p$ is not an equivalence. One can take $\mathcal{C} = \mathrm{Perf}(B)$ for a nonconnective E_1 - $H\mathbb{F}_p$ -algebra B . The two invariants are not naturally equivalent on all small $\mathbb{Z}$ -linear stable categories, even after p -completion.*
- (iv) *At a fixed prime and every height $n \geq 2$, the map $L_{T(n)}\alpha_{\mathcal{C}}$ identifies with the localized cyclotomic trace*

$$L_{T(n)}K(\mathcal{C}) \longrightarrow L_{T(n)}\mathrm{TC}(\mathcal{C}).$$

For $\mathcal{C} = \mathrm{Perf}(KU_p^{\wedge})$, the spectra $K_{\mathrm{mot}}^{\mathrm{Sel}}(\mathcal{C})$ and $K^{\mathrm{Sel}}(\mathcal{C})$ are not equivalent, even after p -completion.

The connective comparison follows from Efimov's representation of TC by the fiber of $\mathbf{1} \rightarrow L_{\mathbb{A}^1}^{\mathrm{mot}}\mathbf{1}$ [E–R, Theorem 9.3]. For general inputs, cyclotomic realization maps the invariant represented by this fiber to TC . Theorem 3.2 expresses the fiber of α as the KU -acyclic part of the fiber of this realization map. This also gives the comparison for finite-width c -structures in Corollary 3.9.

The characteristic- p obstruction comes from continuity. Over $\mathbb{F}_p$, let U_{loc} be the universal invariant with values in $\mathrm{Mot}_{\mathbb{F}_p}^{\mathrm{loc}}$, and let $\mathbf{1}_{\mathbb{F}_p}$ be its unit. For every small $\mathbb{F}_p$ -linear stable category $\mathcal{C}$, we have

$$(1.4) \quad \begin{aligned} K_{\mathrm{mot}}^{\mathrm{Sel}}(\mathcal{C})/p &\simeq \mathrm{map}_{\mathrm{Mot}_{\mathbb{F}_p}^{\mathrm{loc}}}(\mathrm{fib}(\mathbf{1}_{\mathbb{F}_p} \rightarrow L_{\mathbb{A}^1}^{\mathrm{mot}}\mathbf{1}_{\mathbb{F}_p}), U_{\mathrm{loc}}(\mathcal{C}))/p, \\ K^{\mathrm{Sel}}(\mathcal{C})/p &\simeq \mathrm{TC}(\mathcal{C})/p. \end{aligned}$$

The functor on the second line extends to a colimit-preserving functor on $\mathrm{Mot}_{\mathbb{F}_p}^{\mathrm{loc}}$, by the cyclotomic module calculation of Clausen–Mathew–Morrow [CMM, Corollary 2.13]. The functor on the first line is represented by the desuspension of the Moore cofiber of the localization fiber. We prove that this representing object is not compact. The realization theorem of Ramzi–Sosnilo–Winges transfers this failure to a coproduct of small $\mathbb{F}_p$ -linear categories. A localization sequence then produces the single-algebra counterexample. This existence argument proves failure of the specified comparison and of any natural equivalence of the functors. The periodic $KU_p^{\wedge}$ example gives the stronger conclusion that the two value spectra at one input are not equivalent; it lies outside the $\mathbb{Z}$ -linear setting.

Section 3 constructs the comparison from the quotient, and Section 4 proves the arithmetic statements. Sections 5–7 establish the characteristic- p obstruction. Section 8 gives the chromatic calculation, and Appendix A proves the single-algebra realization used in part (iii).

2. LOCALIZING MOTIVES AND CYCLOTOMIC REALIZATION

2.1. Conventions. All stable categories are ∞ -categories. We work with small idempotent-complete stable categories; an invariant of a small stable category is understood to mean the invariant of its idempotent completion. We fix nested universes large enough for the presentable categories and the small categorical realizations used below.

A functor on small stable categories is *localizing* if it takes the zero category to zero and Karoubi localization sequences to fiber sequences. It is *finitary* if it also preserves filtered colimits. Mapping spectra are denoted by map , and internal mapping objects by $\underline{\mathrm{Hom}}$. All tensor products are derived. Presentable stable categories carry their canonical action of spectra.

For an object X of a stable category, put

$$X/p = \mathrm{cofib}(p : X \rightarrow X).$$

For spectra, $X_p^\wedge$ denotes derived p -completion, namely localization at the finite Moore spectrum $\mathbb{S}/p$. This localization is exact, and a map is a p -complete equivalence precisely when it is an equivalence modulo p . Tensoring with $H\mathbb{F}_p$ will always be written explicitly.

Throughout, k denotes a discrete commutative ring and Hk its Eilenberg–Mac Lane commutative ring spectrum. We use R for a connective E_1 -ring spectrum and B for an arbitrary E_1 -ring spectrum. A k -linear category is a module over $\mathrm{Perf}(Hk)$ in $\mathrm{Cat}^{\mathrm{perf}}$, and relative tensor products of such categories are taken over $\mathrm{Perf}(Hk)$. We write $\mathrm{Perf}(k)$ for $\mathrm{Perf}(Hk)$. For ring spectra, $K(B)$, $\mathrm{THH}(B)$, and $\mathrm{TC}(B)$ mean the corresponding invariants of $\mathrm{Perf}(B)$; for discrete rings these are evaluated on Hk . The dg and Hk -linear Morita models are equivalent; see [Shi, Theorem 1.1], [Toë, Sections 6–7], and [BGT, Sections 2–4].

The integral height-one localization L_{KU} of spectra is smashing. Thus

$$(2.1) \quad \begin{aligned} L_{KU}X &\simeq (L_{KU}\mathbb{S}) \otimes X & (X \in \mathrm{Sp}), \\ (L_{KU}\mathbb{S}) \otimes \mathrm{fib}(\mathbb{S} \rightarrow L_{KU}\mathbb{S}) &= 0, \\ (L_{KU}\mathbb{S}) \otimes H\mathbb{F}_p &= 0. \end{aligned}$$

On motives and other presentable stable categories, we use the spectral action $(L_{KU}\mathbb{S}) \otimes -$. The first assertion follows from height-one localization [Rav]; see also [LMMT, Section 2.2]. For the last, $H\mathbb{F}_p$ is rationally trivial and its positive-height periodic homotopy vanishes because it is bounded above [LMMT, Lemma 2.2(ii),(viii)]. At primes other than p , its Moore reductions vanish as well.

2.2. Relative localizing motives. For a discrete commutative ring k , let $D(k) \simeq \mathrm{Mod}_{Hk}$ be its derived category of modules, and write

$$\mathrm{Mot}_k^{\mathrm{loc}} = \mathrm{Mot}_{D(k)}^{\mathrm{loc}}.$$

The universal finitary localizing invariant of small k -linear stable categories is

$$U_{\mathrm{loc}} : \mathrm{Cat}_k^{\mathrm{perf}} \longrightarrow \mathrm{Mot}_k^{\mathrm{loc}}, \quad \mathbf{1}_k = U_{\mathrm{loc}}(\mathrm{Perf}(k)).$$

The base of U_{loc} is specified by its source and target. We use the same notation for its continuous extension to dualizable presentable categories, as in [E–R, Section 1.6]. The categories of finitary localizing motives are those of [BGT, E–R].

Proposition 2.1. *The categories $\mathrm{Mot}_k^{\mathrm{loc}}$ and $\mathrm{Mot}_k^{\mathrm{loc}}$ have the following properties.*

- (i) *They are presentable stable symmetric monoidal categories with compact unit, and*

$$\begin{aligned} \mathrm{map}_{\mathrm{Mot}_k^{\mathrm{loc}}}(\mathbf{1}_k, U_{\mathrm{loc}}(\mathcal{C})) &\simeq K(\mathcal{C}), \\ \mathrm{End}_{\mathrm{Mot}_k^{\mathrm{loc}}}(\mathbf{1}_k) &\simeq K(k). \end{aligned}$$

The analogous formulas hold for $\mathrm{Mot}_k^{\mathrm{loc}}$.

- (ii) *They are rigid. The pairing*

$$(X, Y) \longmapsto \mathrm{map}_{\mathrm{Mot}_k^{\mathrm{loc}}}(\mathbf{1}_k, X \otimes Y)$$

exhibits $\mathrm{Mot}_k^{\mathrm{loc}}$ as its own dual in $\mathrm{Pr}_{\mathrm{st}}^{\mathrm{L}}$. In particular, an object X is zero if this pairing vanishes for every Y .

(iii) *Extension of scalars gives a symmetric monoidal adjunction*

$$f_k^* : \mathrm{Mot}^{\mathrm{loc}} \rightleftarrows \mathrm{Mot}_k^{\mathrm{loc}} : f_{k*}.$$

Both adjoints preserve colimits, and $f_{k}U_{\mathrm{loc}}(\mathcal{C})$ is the motive of the underlying stable category of $\mathcal{C}$.*

Proof. Corepresentability of nonconnective K -theory is [BGT, Theorem 9.8]; its relative form and compactness of the unit are part of [E–R, Sections 1 and 3]. Rigidity and the duality pairing follow from [E–R, Theorem 3.1], with base Sp or Mod_{Hk} . The adjunction and continuity of its right adjoint are [E–R, Proposition 5.1]. $\square$

Mapping out of $\mathbf{1}_k$ is exact and preserves filtered colimits by compactness. It therefore preserves arbitrary coproducts and hence all colimits. In particular, for $V \in \mathrm{Sp}$ it is spectrally linear:

$$(2.2) \quad \mathrm{map}_{\mathrm{Mot}_k^{\mathrm{loc}}}(\mathbf{1}_k, V \otimes X) \simeq V \otimes \mathrm{map}_{\mathrm{Mot}_k^{\mathrm{loc}}}(\mathbf{1}_k, X).$$

The same argument applies to mapping out of $\mathbf{1}$ in $\mathrm{Mot}^{\mathrm{loc}}$.

2.3. Polynomial localization and realization. Write CycSp for the Nikolaus–Scholze category of cyclotomic spectra. For every $X \in \mathrm{CycSp}$, we use

$$(2.3) \quad \mathrm{TC}(X) = \mathrm{map}_{\mathrm{CycSp}}(\mathbf{1}_{\mathrm{CycSp}}, X).$$

Thus $\mathrm{TC}(\mathcal{C}) = \mathrm{TC}(\mathrm{THH}(\mathcal{C}))$, and TC is exact.

We use the simplicial flat polynomial ring spectrum $\mathbb{S}[\Delta^\bullet]$ of [E–R, equation (8.1)], with

$$\mathbb{S}[\Delta^m] = \mathbb{S}[x_1, \dots, x_m], \quad \mathbb{S}[t] = \Sigma_+^\infty \mathbb{N}.$$

The two faces in degree one send x to 0 and 1. For a discrete commutative ring k , the simplicial polynomial ring

$$k[\Delta^m] = k[t_0, \dots, t_m]/(t_0 + \dots + t_m - 1)$$

satisfies the levelwise equivalence of ring spectra

$$H(k[\Delta^\bullet]) \simeq Hk \otimes \mathbb{S}[\Delta^\bullet].$$

Proposition 2.2. (i) *Polynomial localization is smashing, with*

$$\begin{aligned} L_{\mathbb{A}^1}^{\mathrm{mot}} X &\simeq (L_{\mathbb{A}^1}^{\mathrm{mot}} \mathbf{1}) \otimes X, \\ (L_{\mathbb{A}^1}^{\mathrm{mot}} \mathbf{1}) \otimes (L_{\mathbb{A}^1}^{\mathrm{mot}} \mathbf{1}) &\simeq L_{\mathbb{A}^1}^{\mathrm{mot}} \mathbf{1}, \\ L_{\mathbb{A}^1}^{\mathrm{mot}} \mathbf{1} &\simeq |U_{\mathrm{loc}}(\mathrm{Perf}(\mathbb{S}[\Delta^\bullet]))|. \end{aligned}$$

Over k , the corresponding identifications are

$$\begin{aligned} f_k^*(L_{\mathbb{A}^1}^{\mathrm{mot}} \mathbf{1}) &\simeq L_{\mathbb{A}^1}^{\mathrm{mot}} \mathbf{1}_k, \\ L_{\mathbb{A}^1}^{\mathrm{mot}} \mathbf{1}_k &\simeq |U_{\mathrm{loc}}(\mathrm{Perf}(k[\Delta^\bullet]))|. \end{aligned}$$

Extension of scalars carries the unit map $\mathbf{1} \rightarrow L_{\mathbb{A}^1}^{\mathrm{mot}} \mathbf{1}$ and its fiber to their relative counterparts.

(ii) *Cyclotomic THH induces colimit-preserving symmetric monoidal functors*

$$\begin{aligned} \mathrm{THH}^{\mathrm{loc}} : \mathrm{Mot}^{\mathrm{loc}} &\longrightarrow \mathrm{CycSp}, \\ \mathrm{THH}_k^{\mathrm{loc}} : \mathrm{Mot}_k^{\mathrm{loc}} &\longrightarrow \mathrm{Mod}_{\mathrm{THH}(k)}(\mathrm{CycSp}), \end{aligned}$$

with

$$\begin{aligned} \mathrm{THH}^{\mathrm{loc}}(L_{\mathbb{A}^1}^{\mathrm{mot}} \mathbf{1}) &= 0, \\ \mathrm{THH}^{\mathrm{loc}}(\mathrm{fib}(\mathbf{1} \rightarrow L_{\mathbb{A}^1}^{\mathrm{mot}} \mathbf{1})) &\simeq \mathbf{1}_{\mathrm{CycSp}}, \\ \mathrm{THH}_k^{\mathrm{loc}}(L_{\mathbb{A}^1}^{\mathrm{mot}} \mathbf{1}_k) &= 0, \\ \mathrm{THH}_k^{\mathrm{loc}}(\mathrm{fib}(\mathbf{1}_k \rightarrow L_{\mathbb{A}^1}^{\mathrm{mot}} \mathbf{1}_k)) &\simeq \mathrm{THH}(k). \end{aligned}$$

(iii) For every connective E_1 -ring spectrum R , realization induces an equivalence

$$\mathrm{map}_{\mathrm{Mot}^{\mathrm{loc}}}(\mathrm{fib}(\mathbf{1} \rightarrow L_{\mathbb{A}^1}^{\mathrm{mot}} \mathbf{1}), U_{\mathrm{loc}}(\mathrm{Perf}(R))) \longrightarrow \mathrm{TC}(R),$$

compatible with the maps from $K(R)$.

Proof. Part (i) is [E–R, Theorem 8.1]. Cyclotomic THH is a finitary localizing invariant, and its monoidal structure gives the realizations in (ii), retaining the $\mathrm{THH}(k)$ -action over k . The vanishing of $\mathrm{THH}^{\mathrm{loc}}(L_{\mathbb{A}^1}^{\mathrm{mot}} \mathbf{1})$ is proved in [E–R, Section 9.2, proof of Theorem 9.3]: the realization $|\mathrm{THH}(\mathbb{S}[\Delta^\bullet])|$ is a unital ring spectrum in which the two degree-one faces identify 0 and 1 in π_0 . The relative vanishing follows by scalar extension. Applying realization to the fiber sequence of $\mathbf{1} \rightarrow L_{\mathbb{A}^1}^{\mathrm{mot}} \mathbf{1}$ gives the remaining identifications in (ii). Part (iii) is [E–R, Theorem 9.3]. $\square$

3. THE MOTIVIC QUOTIENT AND ITS COMPARISON

Definition 3.1. For a small stable category $\mathcal{C}$, define

$$\mathrm{TC}_{\mathrm{mot}}(\mathcal{C}) = \mathrm{map}_{\mathrm{Mot}^{\mathrm{loc}}}(\mathrm{fib}(\mathbf{1} \rightarrow L_{\mathbb{A}^1}^{\mathrm{mot}} \mathbf{1}), U_{\mathrm{loc}}(\mathcal{C})).$$

Applying cyclotomic realization to maps out of this fiber gives

$$(3.1) \quad \gamma_{\mathcal{C}} : \mathrm{TC}_{\mathrm{mot}}(\mathcal{C}) \longrightarrow \mathrm{TC}(\mathcal{C}).$$

The map from the fiber to $\mathbf{1}$ becomes an equivalence under realization. Consequently, the composite from $K(\mathcal{C})$ is the cyclotomic trace. The fiber sequence

$$(3.2) \quad \mathrm{fib}(\gamma_{\mathcal{C}}) \longrightarrow \mathrm{TC}_{\mathrm{mot}}(\mathcal{C}) \xrightarrow{\gamma_{\mathcal{C}}} \mathrm{TC}(\mathcal{C})$$

determines the error in the Selmer comparison.

Theorem 3.2 (Comparison). *There is a natural transformation $\alpha : K_{\mathrm{mot}}^{\mathrm{Sel}} \rightarrow K^{\mathrm{Sel}}$ and natural equivalences*

$$(3.3) \quad K_{\mathrm{mot}}^{\mathrm{Sel}}(\mathcal{C}) \simeq L_{KU} K(\mathcal{C}) \times_{L_{KU} \mathrm{TC}_{\mathrm{mot}}(\mathcal{C})} \mathrm{TC}_{\mathrm{mot}}(\mathcal{C}),$$

$$(3.4) \quad \mathrm{fib}(\alpha_{\mathcal{C}}) \simeq \mathrm{fib}(\mathrm{fib}(\gamma_{\mathcal{C}}) \longrightarrow L_{KU} \mathrm{fib}(\gamma_{\mathcal{C}})).$$

Under the pullback description, α is induced by γ and the identity of $L_{KU} K$. Both $L_{KU} \alpha_{\mathcal{C}}$ and the rationalization of $\alpha_{\mathcal{C}}$ are equivalences.

We compute the coreflector onto the ideal killed by q . This gives a cofiber formula for $K_{\mathrm{mot}}^{\mathrm{Sel}}$; a stable gluing identity then yields (3.3). Comparing the two pullbacks computes the fiber of α .

3.1. The localizing ideal.

Lemma 3.3. *The localizing tensor ideal generated by the fiber of (1.2) is*

$$\mathcal{I} = \{Y \in \mathrm{Mot}^{\mathrm{loc}} : L_{\mathbb{A}^1}^{\mathrm{mot}} Y \simeq Y \text{ and } (L_{KU} \mathbb{S}) \otimes Y = 0\}.$$

The quotient $q : \mathrm{Mot}^{\mathrm{loc}} \rightarrow \mathrm{Mot}^{\mathrm{loc}} / \mathcal{I}$ is an accessible symmetric monoidal localization.

Proof. The fiber of (1.2) satisfies both conditions. They are preserved under colimits, retracts, and tensoring with arbitrary objects, since both localizations are smashing. Conversely, tensoring (1.2) with an object Y satisfying these conditions gives $Y \rightarrow 0$. Thus Y is the tensor product of that fiber with Y , and belongs to the ideal it generates.

Choose a small family $\{G_\lambda\}$ generating $\mathrm{Mot}^{\mathrm{loc}}$ under colimits. The tensor products of these generators with the fiber of (1.2) generate $\mathcal{I}$ as a localizing subcategory. Hence the localization is accessible. Its tensor-ideal kernel gives the symmetric monoidal structure on the quotient. $\square$

Lemma 3.4. *The inclusion $i : \mathcal{I} \hookrightarrow \text{Mot}^{\text{loc}}$ has a right adjoint $i^!$, with*

$$ii^! X \simeq \text{fib}\left(\underline{\text{Hom}}(L_{\mathbb{A}^1}^{\text{mot}} \mathbf{1}, X) \longrightarrow (L_{KU} \mathbb{S}) \otimes \underline{\text{Hom}}(L_{\mathbb{A}^1}^{\text{mot}} \mathbf{1}, X)\right).$$

Here the internal Hom is taken in Mot^{loc} . If j is the right adjoint of q , then

$$(3.5) \quad jqX \simeq \text{cofib}(ii^! X \longrightarrow X).$$

Proof. Precomposition with multiplication makes $\underline{\text{Hom}}(L_{\mathbb{A}^1}^{\text{mot}} \mathbf{1}, X)$ a module over the localized unit. Modules over an idempotent algebra are its local objects, so this internal Hom is polynomially local. The fiber displayed in the statement is therefore polynomially local and annihilated by tensoring with $L_{KU} \mathbb{S}$; it belongs to $\mathcal{I}$.

For $Y \in \mathcal{I}$, maps from Y to the second term in that fiber vanish. The internal Hom adjunction then gives

$$\text{map}(Y, ii^! X) \simeq \text{map}((L_{\mathbb{A}^1}^{\text{mot}} \mathbf{1}) \otimes Y, X) \simeq \text{map}(Y, X).$$

These equivalences are natural. Their counit is the composite of the fiber inclusion with evaluation along $\mathbf{1} \rightarrow L_{\mathbb{A}^1}^{\text{mot}} \mathbf{1}$. Its cofiber is right orthogonal to $\mathcal{I}$, proving (3.5). $\square$

Proposition 3.5. *There are natural fiber and cofiber sequences*

$$(3.6) \quad \text{map}_{\text{Mot}^{\text{loc}}}(L_{\mathbb{A}^1}^{\text{mot}} \mathbf{1}, U_{\text{loc}}(\mathcal{C})) \longrightarrow K(\mathcal{C}) \longrightarrow \text{TC}_{\text{mot}}(\mathcal{C})$$

and

$$(3.7) \quad \text{fib}\left(\text{map}_{\text{Mot}^{\text{loc}}}(L_{\mathbb{A}^1}^{\text{mot}} \mathbf{1}, U_{\text{loc}}(\mathcal{C})) \longrightarrow L_{KU} \text{map}_{\text{Mot}^{\text{loc}}}(L_{\mathbb{A}^1}^{\text{mot}} \mathbf{1}, U_{\text{loc}}(\mathcal{C}))\right) \longrightarrow K(\mathcal{C}) \longrightarrow K_{\text{mot}}^{\text{Sel}}(\mathcal{C}).$$

The mapping-spectrum functor in (3.6), TC_{mot} , and $K_{\text{mot}}^{\text{Sel}}$ are localizing invariants.

Proof. Mapping the fiber sequence of $\mathbf{1} \rightarrow L_{\mathbb{A}^1}^{\text{mot}} \mathbf{1}$ into $U_{\text{loc}}(\mathcal{C})$ gives (3.6). The quotient adjunction identifies $K_{\text{mot}}^{\text{Sel}}(\mathcal{C})$ with $\text{map}(\mathbf{1}, jqU_{\text{loc}}(\mathcal{C}))$. Apply $\text{map}(\mathbf{1}, -)$ to (3.5) and use the formula for $ii^!$. Exactness, the internal Hom adjunction, and the spectral linearity (2.2) identify the first term with the fiber displayed in (3.7). Mapping spectra are exact in the second variable, so these functors take localization sequences to fiber sequences. $\square$

Remark 3.6. The spectral linearity in this calculation comes from the compact unit of Mot^{loc} .

3.2. Gluing.

Lemma 3.7. *Let L be an exact localization of spectra, and let $F \rightarrow E \rightarrow T$ be a fiber sequence. Then*

$$\text{cofib}(\text{fib}(F \rightarrow LF) \longrightarrow E) \simeq LE \times_{LT} T$$

naturally in the fiber sequence.

Proof. The left side is the pushout $P = E \amalg_F LF$. Thus $LF \rightarrow P \rightarrow T$ is a fiber sequence, and $LP \simeq LE$. The resulting map $P \rightarrow LE \times_{LT} T$ is a map over T which identifies the fibers with LF , hence is an equivalence. $\square$

Proof of Theorem 3.2. Apply Lemma 3.7 with $L = L_{KU}$ to (3.6). The cofiber formula (3.7) then gives (3.3). The realization γ , together with the identity of $L_{KU} K$, defines α .

Write each pullback as the fiber of the map from the direct sum of its two upper terms to the lower term. On taking the fiber of the comparison, the common $L_{KU} K(\mathcal{C})$ term contributes zero. The remaining fiber is

$$\text{fib}(\text{fib}(\gamma_{\mathcal{C}}) \longrightarrow L_{KU} \text{fib}(\gamma_{\mathcal{C}})),$$

which proves (3.4). Tensoring with $L_{KU} \mathbb{S}$ or with $H\mathbb{Q}$ kills this fiber. $\square$

Corollary 3.8 (Connective comparison). *For every connective E_1 -ring spectrum R , the map $\alpha_{\text{Perf}(R)}$ is an integral equivalence.*

Proof. Proposition 2.2(iii) makes $\gamma_{\text{Perf}(R)}$ an equivalence. Apply (3.4). $\square$

Corollary 3.9 (Finite-width comparison). *The map $\alpha_{\mathcal{C}}$ is an equivalence if $\mathcal{C}$ admits a finite-width c -structure in the sense of [LS, Definitions 3.1 and 3.4].*

Proof. The fiber of α is a localizing invariant, by Proposition 3.5 and exactness of TC and L_{KU} . If $\mathcal{C}$ has width at most n , [LS, Proposition 5.3] gives a connective ring spectrum $R_{\mathcal{C}}$ such that

$$\Sigma^{\max(n,1)} E(\mathcal{C}) \simeq E(\text{Perf}(R_{\mathcal{C}}))$$

for every localizing invariant E . Apply this to $E = \text{fib}(\alpha)$. Corollary 3.8 makes the right-hand side zero, and hence $\text{fib}(\alpha_{\mathcal{C}}) = 0$. $\square$

4. RELATIVE MOTIVES AND ARITHMETIC COMPARISON

Theorem 4.1 (Prime-inverted comparison). *Let p be a prime and $\mathcal{C}$ a small $\mathbb{Z}[1/p]$ -linear stable category. There are natural equivalences*

$$K_{\text{mot}}^{\text{Sel}}(\mathcal{C})_p^{\wedge} \simeq K^{\text{Sel}}(\mathcal{C})_p^{\wedge} \simeq (L_{KU}K(\mathcal{C}))_p^{\wedge},$$

the first induced by $\alpha_{\mathcal{C}}$.

4.1. Restriction to a linear base. For a discrete commutative ring k and a k -linear category $\mathcal{C}$, the scalar-extension adjunction of Proposition 2.1(iii) gives

$$(4.1) \quad \begin{aligned} \text{map}_{\text{Mot}^{\text{loc}}} (L_{\mathbb{A}^1}^{\text{mot}} \mathbf{1}, U_{\text{loc}}(\mathcal{C})) \\ \simeq \text{map}_{\text{Mot}_k^{\text{loc}}} (L_{\mathbb{A}^1}^{\text{mot}} \mathbf{1}_k, U_{\text{loc}}(\mathcal{C})). \end{aligned}$$

$$(4.2) \quad \text{TC}_{\text{mot}}(\mathcal{C}) \simeq \text{map}_{\text{Mot}_k^{\text{loc}}} (\text{fib}(\mathbf{1}_k \rightarrow L_{\mathbb{A}^1}^{\text{mot}} \mathbf{1}_k), U_{\text{loc}}(\mathcal{C})).$$

These equivalences identify the fiber sequence (3.6) with the mapping-spectrum sequence induced by

$$\text{fib}(\mathbf{1}_k \rightarrow L_{\mathbb{A}^1}^{\text{mot}} \mathbf{1}_k) \longrightarrow \mathbf{1}_k \longrightarrow L_{\mathbb{A}^1}^{\text{mot}} \mathbf{1}_k.$$

The free-module adjunction gives

$$\text{map}_{\text{Mod}_{\text{THH}(k)}(\text{CycSp})} (\text{THH}(k), \text{THH}(\mathcal{C})) \simeq \text{TC}(\mathcal{C}).$$

Together with compatibility of realization and scalar extension, this identifies the relative realization map with $\gamma_{\mathcal{C}}$. Thus (4.1) and (4.2) compute the restriction of the absolute construction to k -linear categories.

Every mapping spectrum in $\text{Mot}_k^{\text{loc}}$ is naturally a module over $\text{End}(\mathbf{1}_k) \simeq K(k)$. The cyclotomic $\text{THH}(k)$ -module structure on $\text{THH}(\mathcal{C})$ similarly makes $\text{TC}(\mathcal{C})$ a $\text{TC}(k)$ -module, and hence a $K(k)$ -module through the cyclotomic trace.

4.2. Coefficients away from the characteristic.

Lemma 4.2. *For every prime p , one has*

$$\text{fib}(\mathbf{1}_{\mathbb{Z}[1/p]} \longrightarrow L_{\mathbb{A}^1}^{\text{mot}} \mathbf{1}_{\mathbb{Z}[1/p]})/p = 0$$

in $\text{Mot}_{\mathbb{Z}[1/p]}^{\text{loc}}$.

Proof. Consider the constant map

$$\mathbf{1}_{\mathbb{Z}[1/p]} \longrightarrow U_{\text{loc}}(\text{Perf}(\mathbb{Z}[1/p][t])).$$

For every small $\mathbb{Z}[1/p]$ -linear stable category $\mathcal{D}$, [Tab, Theorem 1.2(i)] gives

$$K(\mathcal{D})/p \xrightarrow{\sim} K(\mathcal{D} \otimes_{\text{Perf}(\mathbb{Z}[1/p])} \text{Perf}(\mathbb{Z}[1/p][t]))/p.$$

Here we apply the theorem to a dg Morita model of $\mathcal{D}$, using the comparison in Section 2; the flatness of $\mathbb{Z}[1/p][t]$ over $\mathbb{Z}[1/p]$ identifies its polynomial extension with the displayed derived tensor product. By monoidality and (2.2), the displayed equivalence is obtained from the constant map modulo p by tensoring with $U_{\text{loc}}(\mathcal{D})$ and mapping out of the unit. Both resulting

functors preserve colimits in the tensor factor. Since these motives generate $\text{Mot}_{\mathbb{Z}[1/p]}^{\text{loc}}$, the same comparison is an equivalence after tensoring with any motive and mapping out of the unit. The nondegenerate pairing of Proposition 2.1(ii), applied to the cofiber, shows that the constant map is itself an equivalence modulo p .

Tensoring this equivalence with polynomial motives and iterating the polynomial extension shows that the constant maps induce equivalences

$$\mathbf{1}_{\mathbb{Z}[1/p]}/p \longrightarrow U_{\text{loc}}(\text{Perf}(\mathbb{Z}[1/p][\Delta^m]))/p \quad (m \geq 0).$$

These maps form a map of simplicial objects. Geometric realization commutes with reduction modulo p , so Proposition 2.2(i) gives

$$\mathbf{1}_{\mathbb{Z}[1/p]}/p \simeq (L_{\mathbb{A}^1}^{\text{mot}} \mathbf{1}_{\mathbb{Z}[1/p]})/p.$$

The fiber of this map vanishes, proving the lemma. $\square$

Proof of Theorem 4.1. For objects W, Z of a stable category, finite Moore duality gives

$$(4.3) \quad \text{map}(W, Z)/p \simeq \text{map}(\Sigma^{-1}(W/p), Z).$$

Indeed, the dual of $\mathbb{S}/p$ is $\Sigma^{-1}\mathbb{S}/p$. Lemma 4.2 and (4.2) now give $\text{TC}_{\text{mot}}(\mathcal{C})/p = 0$.

Since p is invertible in $\mathbb{Z}[1/p]$, the $\text{THH}(\mathbb{Z}[1/p])$ -module $\text{THH}(\mathcal{C})$ has p -invertible underlying spectrum. The forgetful functor from cyclotomic spectra is conservative, so multiplication by p is an equivalence of cyclotomic spectra. Applying TC gives $\text{TC}(\mathcal{C})/p = 0$.

Reduction modulo p is exact. The pullbacks (3.3) and (1.1) therefore both reduce to $(L_{KU}K(\mathcal{C}))/p$, and $\alpha_{\mathcal{C}}/p$ identifies with its identity map. Derived p -completion proves the theorem. $\square$

Corollary 4.3 (Rational-linear comparison). *For every small $\mathbb{Q}$ -linear stable category $\mathcal{C}$, the map $\alpha_{\mathcal{C}}$ is an integral equivalence.*

Proof. Theorem 4.1 shows that multiplication by every prime is invertible on $\text{fib}(\alpha_{\mathcal{C}})$. This fiber is therefore rational. Its rationalization vanishes by Theorem 3.2, so the fiber itself vanishes. $\square$

Corollary 4.4 (Support of the arithmetic defect). *Let p be a prime and $\mathcal{C}$ a small $\mathbb{Z}$ -linear stable category, and put*

$$\mathcal{C}_{p\text{-tors}} = \mathcal{C} \otimes_{\text{Perf}(\mathbb{Z})} \text{Perf}_{(p)}(\mathbb{Z}),$$

where $\text{Perf}_{(p)}(\mathbb{Z})$ consists of the perfect complexes annihilated after inverting p . Then the inclusion induces an equivalence

$$\text{fib}(\alpha_{\mathcal{C}_{p\text{-tors}}})_p^{\wedge} \xrightarrow{\sim} \text{fib}(\alpha_{\mathcal{C}})_p^{\wedge}.$$

Proof. Tensoring the localization sequence

$$\text{Perf}_{(p)}(\mathbb{Z}) \longrightarrow \text{Perf}(\mathbb{Z}) \longrightarrow \text{Perf}(\mathbb{Z}[1/p])$$

with $\mathcal{C}$ over $\text{Perf}(\mathbb{Z})$ gives a Karoubi localization sequence. The tensor-localization argument of [BGT, Lemma 9.35] applies in this relative setting by passing to Ind-categories over $\text{Mod}_{H\mathbb{Z}}$, where inversion of p is a smashing localization; see [BGT, Section 5]. Apply the localizing invariant $\text{fib}(\alpha_{(-)})$, followed by derived p -completion. The last category is $\mathbb{Z}[1/p]$ -linear, so its completed fiber vanishes by Theorem 4.1. This gives the asserted equivalence. $\square$

The passage from $\text{Perf}_{(p)}(\mathbb{Z})$ to $\text{Perf}(\mathbb{F}_p)$ by dévissage computes K -theory; a corresponding passage in the proof would require an equivalence of localizing motives.

5. CONTINUITY IN CHARACTERISTIC p

Fix a prime p . For a $\text{THH}(\mathbb{F}_p)$ -module in cyclotomic spectra, TC denotes the TC of its underlying cyclotomic spectrum.

5.1. Reduction of the two Selmer invariants.

Proposition 5.1. *For every small $\mathbb{F}_p$ -linear stable category $\mathcal{C}$, there are natural equivalences*

$$(5.1) \quad K_{\text{mot}}^{\text{Sel}}(\mathcal{C})/p \simeq \text{map}_{\text{Mot}_{\mathbb{F}_p}^{\text{loc}}}(\text{fib}(\mathbf{1}_{\mathbb{F}_p} \rightarrow L_{\mathbb{A}^1}^{\text{mot}} \mathbf{1}_{\mathbb{F}_p}), U_{\text{loc}}(\mathcal{C}))/p.$$

$$(5.2) \quad K^{\text{Sel}}(\mathcal{C})/p \simeq \text{TC}(\mathcal{C})/p.$$

Under these equivalences, $\alpha_{\mathcal{C}}/p$ is the realization map.

Lemma 5.2. *If V is a $K(\mathbb{F}_p)$ -module spectrum, then V/p is naturally an $H\mathbb{F}_p$ -module spectrum. In particular, $L_{KU}(V/p) = 0$.*

Proof. Quillen's calculation [Qui-F] gives

$$K_0(\mathbb{F}_p) = \mathbb{Z}, \quad K_{2i}(\mathbb{F}_p) = 0, \quad K_{2i-1}(\mathbb{F}_p) = \mathbb{Z}/(p^i - 1) \quad (i \geq 1).$$

The negative groups vanish because $\mathbb{F}_p$ is regular. Since p is prime to $p^i - 1$, the truncation map $K(\mathbb{F}_p) \rightarrow H\mathbb{Z}$ induces an equivalence $K(\mathbb{F}_p)/p \simeq H\mathbb{F}_p$ of $K(\mathbb{F}_p)$ -modules. Hence

$$V/p \simeq V \otimes_{K(\mathbb{F}_p)} (K(\mathbb{F}_p)/p) \simeq V \otimes_{K(\mathbb{F}_p)} H\mathbb{F}_p.$$

This gives the $H\mathbb{F}_p$ -module structure, and (2.1) gives the vanishing. $\square$

Proof of Proposition 5.1. The spectra in (4.1) and (4.2), together with $K(\mathcal{C})$ and $\text{TC}(\mathcal{C})$, are $K(\mathbb{F}_p)$ -modules. Lemma 5.2 makes their reductions modulo p L_{KU} -acyclic. After reduction modulo p , the first terms of the cofiber sequences (3.7) and (3.6) therefore identify. Their final terms identify as well, and (4.2) gives (5.1).

Since L_{KU} is smashing, $(L_{KU}V)/p \simeq L_{KU}(V/p)$. Reduction modulo p also preserves finite pullbacks. Applying it to (1.1), the two L_{KU} -local terms vanish and give (5.2). The compatibility of realization with the trace identifies $\alpha_{\mathcal{C}}/p$ with the resulting realization map. $\square$

5.2. Continuity of cyclotomic realization.

Corollary 5.3. *The functor*

$$\text{Mot}_{\mathbb{F}_p}^{\text{loc}} \longrightarrow \text{Sp}, \quad X \longmapsto \text{TC}(\text{THH}_{\mathbb{F}_p}^{\text{loc}}(X))/p$$

preserves all colimits. In particular, $\text{TC}(-)/p$ is a finitary localizing invariant of all small $\mathbb{F}_p$ -linear stable categories.

Lemma 5.4. *The functor*

$$\text{Mod}_{\text{THH}(\mathbb{F}_p)}(\text{CycSp}) \longrightarrow \text{Sp}, \quad X \longmapsto \text{TC}(X)/p$$

preserves all small colimits.

Proof. Let $\text{triv} : \text{Sp} \rightarrow \text{CycSp}$ be the colimit-preserving symmetric monoidal functor left adjoint to TC . It equips a spectrum with the trivial circle action and the canonical cyclotomic Frobenius; see [NS, Proposition IV.4.14] or [CMM, Definition 2.9]. The calculation $\tau_{\geq 0} \text{TC}(\mathbb{F}_p) \simeq H\mathbb{Z}_p$ gives a map of commutative ring spectra $H\mathbb{Z}_p \rightarrow \text{TC}(\mathbb{F}_p)$. Its adjoint is a map of commutative cyclotomic algebras

$$(5.3) \quad (H\mathbb{Z}_p)^{\text{triv}} \longrightarrow \text{THH}(\mathbb{F}_p).$$

This map is described in [NS, Section IV.4, after Proposition IV.4.14, and Corollary IV.4.16] and recalled after [CMM, Lemma 2.10].

Restricting scalars along (5.3) makes every $X \in \text{Mod}_{\text{THH}(\mathbb{F}_p)}(\text{CycSp})$ an $(H\mathbb{Z}_p)^{\text{triv}}$ -module. Since triv is exact, its Moore cofiber is

$$X/p \simeq X \otimes_{(H\mathbb{Z}_p)^{\text{triv}}} (H\mathbb{F}_p)^{\text{triv}}.$$

Thus $X \mapsto X/p$ takes values in $\text{Mod}_{(H\mathbb{F}_p)^{\text{triv}}}(\text{CycSp})$ and preserves colimits: restriction of scalars preserves colimits of modules, and extension of scalars is a left adjoint.

By [CMM, Corollary 2.13], the functor

$$\mathrm{TC} : \mathrm{Mod}_{(H\mathbb{F}_p)^{\mathrm{triv}}}(\mathrm{CycSp}) \longrightarrow \mathrm{Sp}$$

preserves all colimits. Indeed, [CMM, Proposition 2.12] gives on this module category a natural fiber sequence

$$\mathrm{TC}(Y) \longrightarrow Y \longrightarrow Y_{h\mathbb{T}},$$

where the middle term is the underlying spectrum. Both of the latter functors preserve colimits. Finally, exactness of TC identifies $\mathrm{TC}(X/p)$ with $\mathrm{TC}(X)/p$, proving the lemma. $\square$

Proof of Corollary 5.3. By Proposition 2.2, the realization $\mathrm{THH}_{\mathbb{F}_p}^{\mathrm{loc}}$ preserves colimits and takes values in $\mathrm{Mod}_{\mathrm{THH}(\mathbb{F}_p)}(\mathrm{CycSp})$. Apply Lemma 5.4 and then compose with U_{loc} . $\square$

6. A NONCOMPACTNESS THEOREM

Theorem 6.1. *For every prime p , the fiber of*

$$\mathbf{1}_{\mathbb{F}_p}/p \longrightarrow (L_{\mathbb{A}^1}^{\mathrm{mot}} \mathbf{1}_{\mathbb{F}_p})/p$$

is not compact in $\mathrm{Mot}_{\mathbb{F}_p}^{\mathrm{loc}}$.

Lemma 6.2. *Let $\mathcal{E}$ be a rigid presentable stable symmetric monoidal category. Every compact object of $\mathcal{E}$ is dualizable. If F is an E_∞ -ring spectrum for which Mod_F is rigid, then $\mathcal{E} \otimes \mathrm{Mod}_F$ is rigid, and the scalar-extension functor*

$$s : \mathcal{E} \longrightarrow \mathcal{E} \otimes \mathrm{Mod}_F \simeq \mathrm{Mod}_{F \otimes \mathbf{1}}(\mathcal{E})$$

preserves compact objects.

Proof. By [E–I, Proposition 1.12(i)], a compact object x is relatively compact in the $\mathcal{E}$ -module $\mathcal{E}$. Thus the right adjoint of $- \otimes x$ is continuous and $\mathcal{E}$ -linear. An $\mathcal{E}$ -linear endofunctor of $\mathcal{E}$ is tensoring with its value on the unit. The adjunction therefore has the form $- \otimes x \dashv - \otimes x^\vee$, and its unit and counit exhibit $x^\vee$ as a dual.

The multiplication on $\mathcal{E} \otimes \mathrm{Mod}_F$ is the external tensor product of the two multiplication functors, after reordering the factors. Tensoring their continuous bimodule right adjoints gives a continuous bimodule right adjoint to this multiplication. Moreover, the free–forgetful adjunction gives

$$\mathrm{map}(s\mathbf{1}, Z) \simeq \mathrm{map}(\mathbf{1}, \mathrm{Forget} Z).$$

The forgetful functor creates filtered colimits, so $s\mathbf{1}$ is compact. These properties give rigidity; see [E–I, Definition 1.1 and Proposition 1.3]. Replacing $\mathbf{1}$ by a compact x in the same adjunction proves that $s(x)$ is compact. $\square$

Lemma 6.3 (A dualizable idempotent splits the unit). *Let E be an idempotent commutative algebra object in a stable symmetric monoidal category. If E is dualizable, then*

$$\mathbf{1} \simeq E \oplus \mathrm{fib}(\mathbf{1} \rightarrow E).$$

Proof. Idempotence gives $E \otimes \mathrm{fib}(\mathbf{1} \rightarrow E) = 0$. The duality identities exhibit $E^\vee$ as a retract of $E^\vee \otimes E \otimes E^\vee$, so $E^\vee \otimes \mathrm{fib}(\mathbf{1} \rightarrow E) = 0$ as well. Hence

$$\mathrm{map}(E, \Sigma \mathrm{fib}(\mathbf{1} \rightarrow E)) \simeq \mathrm{map}(\mathbf{1}, E^\vee \otimes \Sigma \mathrm{fib}(\mathbf{1} \rightarrow E)) = 0.$$

The connecting map of $\mathrm{fib}(\mathbf{1} \rightarrow E) \rightarrow \mathbf{1} \rightarrow E$ is zero, and the triangle splits. $\square$

Proof of Theorem 6.1. If the fiber in the theorem were compact, the localized unit would become compact and dualizable after extending spectral coefficients to $H\mathbb{F}_p$. Its idempotence would then split the unit. We will show that both summands are nonzero, while the ring of degree-zero endomorphisms of the unit is $\mathbb{F}_p$, giving a contradiction.

Assume that the fiber is compact. Since $\mathbf{1}_{\mathbb{F}_p}/p$ is compact, its cofiber sequence shows that $(L_{\mathbb{A}^1}^{\mathrm{mot}} \mathbf{1}_{\mathbb{F}_p})/p$ is compact. Extend spectral coefficients by

$$s : \mathrm{Mot}_{\mathbb{F}_p}^{\mathrm{loc}} \longrightarrow \mathrm{Mot}_{\mathbb{F}_p}^{\mathrm{loc}} \otimes \mathrm{Mod}_{H\mathbb{F}_p}.$$

Identify the target with the module category

$$\mathrm{Mod}_{H\mathbb{F}_p \otimes \mathbf{1}_{\mathbb{F}_p}}(\mathrm{Mot}_{\mathbb{F}_p}^{\mathrm{loc}}).$$

The underlying motive of $s(X)$ is $H\mathbb{F}_p \otimes X$. Proposition 2.1 and Lemma 6.2 imply that this module category is rigid and that s preserves compact objects. Multiplication by p is nullhomotopic in this category, so

$$\begin{aligned} s((L_{\mathbb{A}^1}^{\mathrm{mot}} \mathbf{1}_{\mathbb{F}_p})/p) &\simeq s(L_{\mathbb{A}^1}^{\mathrm{mot}} \mathbf{1}_{\mathbb{F}_p}) \\ &\oplus \Sigma s(L_{\mathbb{A}^1}^{\mathrm{mot}} \mathbf{1}_{\mathbb{F}_p}). \end{aligned}$$

Thus $s(L_{\mathbb{A}^1}^{\mathrm{mot}} \mathbf{1}_{\mathbb{F}_p})$ is a retract of a compact object and is compact. It is dualizable by Lemma 6.2 and is an idempotent algebra because s is symmetric monoidal. Lemma 6.3, together with exactness of s , gives

$$(6.1) \quad \begin{aligned} s(\mathbf{1}_{\mathbb{F}_p}) &\simeq s(L_{\mathbb{A}^1}^{\mathrm{mot}} \mathbf{1}_{\mathbb{F}_p}) \\ &\oplus s(\mathrm{fib}(\mathbf{1}_{\mathbb{F}_p} \rightarrow L_{\mathbb{A}^1}^{\mathrm{mot}} \mathbf{1}_{\mathbb{F}_p})). \end{aligned}$$

In this module category, the free–forgetful adjunction and (2.2) compute the endomorphisms of the unit as

$$\begin{aligned} \mathrm{End}(s(\mathbf{1}_{\mathbb{F}_p})) &\simeq \mathrm{map}_{\mathrm{Mot}_{\mathbb{F}_p}^{\mathrm{loc}}}(\mathbf{1}_{\mathbb{F}_p}, H\mathbb{F}_p \otimes \mathbf{1}_{\mathbb{F}_p}) \\ &\simeq K(\mathbb{F}_p) \otimes H\mathbb{F}_p. \end{aligned}$$

Both factors are connective, so

$$(6.2) \quad \pi_0 \mathrm{End}(s(\mathbf{1}_{\mathbb{F}_p})) = \pi_0 K(\mathbb{F}_p) \otimes_{\mathbb{Z}} \mathbb{F}_p = \mathbb{F}_p.$$

This ring has no nontrivial idempotents. It remains to show that both summands in (6.1) are nonzero.

The realization formula for the localized unit and preservation of colimits by mapping out of the compact unit give

$$(6.3) \quad \begin{aligned} \mathrm{map}_{\mathrm{Mot}_{\mathbb{F}_p}^{\mathrm{loc}}}(\mathbf{1}_{\mathbb{F}_p}, L_{\mathbb{A}^1}^{\mathrm{mot}} \mathbf{1}_{\mathbb{F}_p}) \\ \simeq |K(\mathbb{F}_p[\Delta^\bullet])| \simeq K(\mathbb{F}_p). \end{aligned}$$

Indeed, the constants define a termwise equivalence from the constant simplicial spectrum $K(\mathbb{F}_p)$, by homotopy invariance for regular rings [Qui–H, Section 6, Theorem 8 and its Corollary]. Nonconnective K -theory agrees with connective K -theory on these rings. Applying the free–forgetful adjunction once more, we obtain in the module category

$$\mathrm{map}(s(\mathbf{1}_{\mathbb{F}_p}), s(L_{\mathbb{A}^1}^{\mathrm{mot}} \mathbf{1}_{\mathbb{F}_p})) \simeq K(\mathbb{F}_p) \otimes H\mathbb{F}_p \neq 0,$$

since this spectrum has $\pi_0 = \mathbb{F}_p$. Thus the first summand of (6.1) is nonzero.

If the second summand were zero, forgetting the module structure would give

$$H\mathbb{F}_p \otimes \mathrm{fib}(\mathbf{1}_{\mathbb{F}_p} \rightarrow L_{\mathbb{A}^1}^{\mathrm{mot}} \mathbf{1}_{\mathbb{F}_p}) = 0.$$

Apply the spectrally linear functor $\mathrm{THH}_{\mathbb{F}_p}^{\mathrm{loc}}$. Proposition 2.2 identifies its value on this fiber with $\mathrm{THH}(\mathbb{F}_p)$, so we would have

$$H\mathbb{F}_p \otimes \mathrm{THH}(\mathbb{F}_p) = 0.$$

But $\mathrm{THH}(\mathbb{F}_p)$ is connective with $\pi_0 \mathrm{THH}(\mathbb{F}_p) = \mathbb{F}_p$, and hence $\pi_0(H\mathbb{F}_p \otimes \mathrm{THH}(\mathbb{F}_p)) = \mathbb{F}_p$. The second summand is therefore nonzero as well.

The splitting (6.1) now gives a nonzero idempotent different from the identity in (6.2), a contradiction. $\square$

Remark 6.4. For a motive X in the source of s , the underlying motive of $s(X)$ is $H\mathbb{F}_p \otimes X$, whereas $s(X/p) \simeq s(X) \oplus \Sigma s(X)$. The proof uses this retract to transfer compactness and the calculation $\pi_0(K(\mathbb{F}_p) \otimes H\mathbb{F}_p) = \mathbb{F}_p$ to exclude a splitting of the unit.

7. FAILURE OF THE COMPARISON ON LINEAR CATEGORIES

Theorem 7.1 (Characteristic- p failure). *For every prime p , there exists a small $\mathbb{F}_p$ -linear stable category $\mathcal{C}$ such that*

$$\alpha_{\mathcal{C}}/p : K_{\text{mot}}^{\text{Sel}}(\mathcal{C})/p \longrightarrow K^{\text{Sel}}(\mathcal{C})/p$$

is not an equivalence. The same is true of $(\alpha_{\mathcal{C}})_p^{\wedge}$. Moreover, the two functors modulo p are not naturally equivalent on all small $\mathbb{F}_p$ -linear stable categories.

By Theorem 6.1, mapping out of the fiber of $\mathbf{1}_{\mathbb{F}_p} \rightarrow L_{\mathbb{A}^1}^{\text{mot}} \mathbf{1}_{\mathbb{F}_p}$, modulo p , fails to preserve some coproduct in $\text{Mot}_{\mathbb{F}_p}^{\text{loc}}$. We realize its summands by small $\mathbb{F}_p$ -linear categories. Their coproduct gives the counterexample because $\text{TC}(\text{THH}_{\mathbb{F}_p}^{\text{loc}}(-))/p$ preserves coproducts by Corollary 5.3.

7.1. Realization by small categories.

Lemma 7.2 (Small relative realization). *Let k be an ordinary commutative ring. For every object $X \in \text{Mot}_k^{\text{loc}}$, there is a small k -linear idempotent-complete stable category $\mathcal{C}$ and an equivalence $U_{\text{loc}}(\mathcal{C}) \simeq X$.*

Proof. We realize X by a presentable category, express its motive as the loop of a small Calkin motive, and realize that loop by the Grayson construction.

The relative localization theorem [RSW, Theorem B.6], applied to $\mathcal{V} = \text{Mod}_{Hk}$, identifies relative motives with the localization of dualizable presentable k -linear categories at motivic equivalences. Its universal invariant is the continuous extension of U_{loc} ; see [E–R, Section 1.6] for the identification of the small and presentable versions. Essential surjectivity gives a dualizable presentable k -linear category $\mathcal{D}$ with

$$X \simeq U_{\text{loc}}(\mathcal{D}).$$

For a sufficiently large uncountable regular cardinal κ , the continuous Calkin construction gives an exact sequence

$$\mathcal{D} \longrightarrow \text{Ind}(\mathcal{D}^{\kappa}) \longrightarrow \text{Ind}(\text{Calk}_{\kappa}^{\text{cont}}(\mathcal{D})).$$

Here $\text{Calk}_{\kappa}^{\text{cont}}(\mathcal{D})$ is the small category of compact objects in the quotient of $\text{Ind}(\mathcal{D}^{\kappa})$ by the image of the continuous Yoneda functor [E–L, Definition 1.60]. The sequence is k -linear by [E–R, Section 1.5]; see also [RSW, Appendix B, discussion following Observation B.7]. For $P \in \text{Perf}(k)$, tensoring with P has the colimit-preserving right adjoint given by tensoring with $P^{\vee}$, so the induced $\text{Perf}(k)$ -action preserves compact objects in the quotient. Thus the Calkin category is k -linear.

The category $\mathcal{D}^{\kappa}$ has countable coproducts. The endofunctor $Z \mapsto \bigoplus_{n \geq 0} Z$ is exact and k -linear, so the Eilenberg swindle gives $U_{\text{loc}}(\mathcal{D}^{\kappa}) = 0$. Applying the continuous invariant to the exact sequence therefore yields

$$(7.1) \quad U_{\text{loc}}(\mathcal{D}) \simeq \Omega U_{\text{loc}}(\text{Calk}_{\kappa}^{\text{cont}}(\mathcal{D})).$$

This is the continuous-extension formula of [E–L, Definition 4.5], in its relative form [E–R, Section 1.6].

The Grayson construction satisfies

$$U_{\text{loc}}(\text{Gr}(\mathcal{A})) \simeq \Omega U_{\text{loc}}(\mathcal{A}).$$

Indeed, Gr is tensoring with a fixed small stable category, and [RSW, Theorem A.10] applies to the localizing invariant $\mathcal{B} \mapsto U_{\text{loc}}(\mathcal{B} \otimes \mathcal{A})$, evaluated at $\text{Perf}(\mathbb{S})$. Taking $\mathcal{C} = \text{Gr}(\text{Calk}_{\kappa}^{\text{cont}}(\mathcal{D}))$ now realizes X . $\square$

7.2. The continuity obstruction. Apply $\mathrm{THH}_{\mathbb{F}_p}^{\mathrm{loc}}$ to maps out of the fiber of $\mathbf{1}_{\mathbb{F}_p} \rightarrow L_{\mathbb{A}^1}^{\mathrm{mot}} \mathbf{1}_{\mathbb{F}_p}$. Proposition 2.2 identifies the realization of this fiber with $\mathrm{THH}(\mathbb{F}_p)$, via its map to $\mathbf{1}_{\mathbb{F}_p}$. We obtain

$$(7.2) \quad \begin{aligned} & \mathrm{map}_{\mathrm{Mot}_{\mathbb{F}_p}^{\mathrm{loc}}} \left(\mathrm{fib}(\mathbf{1}_{\mathbb{F}_p} \rightarrow L_{\mathbb{A}^1}^{\mathrm{mot}} \mathbf{1}_{\mathbb{F}_p}), X \right) / p \\ & \xrightarrow{\gamma_X/p} \mathrm{TC}(\mathrm{THH}_{\mathbb{F}_p}^{\mathrm{loc}}(X)) / p, \quad X \in \mathrm{Mot}_{\mathbb{F}_p}^{\mathrm{loc}}. \end{aligned}$$

The target is identified by the free-forgetful adjunction:

$$\begin{aligned} & \mathrm{map}_{\mathrm{Mod}_{\mathrm{THH}(\mathbb{F}_p)}(\mathrm{CycSp})}(\mathrm{THH}(\mathbb{F}_p), \mathrm{THH}_{\mathbb{F}_p}^{\mathrm{loc}}(X)) \\ & \simeq \mathrm{map}_{\mathrm{CycSp}}(\mathbf{1}_{\mathrm{CycSp}}, \mathrm{THH}_{\mathbb{F}_p}^{\mathrm{loc}}(X)) = \mathrm{TC}(\mathrm{THH}_{\mathbb{F}_p}^{\mathrm{loc}}(X)). \end{aligned}$$

Under (4.2), the value of (7.2) at $X = U_{\mathrm{loc}}(\mathcal{C})$ is $\gamma_{\mathcal{C}}/p$. Proposition 5.1 identifies it with $\alpha_{\mathcal{C}}/p$.

Lemma 7.3. *The functor*

$$\begin{aligned} & \mathrm{Mot}_{\mathbb{F}_p}^{\mathrm{loc}} \longrightarrow \mathrm{Sp}, \\ & X \longmapsto \mathrm{map}_{\mathrm{Mot}_{\mathbb{F}_p}^{\mathrm{loc}}} \left(\mathrm{fib}(\mathbf{1}_{\mathbb{F}_p} \rightarrow L_{\mathbb{A}^1}^{\mathrm{mot}} \mathbf{1}_{\mathbb{F}_p}), X \right) / p \end{aligned}$$

does not preserve all filtered colimits, and does not preserve all small coproducts.

Proof. Finite Moore duality (4.3) identifies the value at X with

$$\mathrm{map}_{\mathrm{Mot}_{\mathbb{F}_p}^{\mathrm{loc}}} \left(\Sigma^{-1} \mathrm{fib}(\mathbf{1}_{\mathbb{F}_p}/p \rightarrow (L_{\mathbb{A}^1}^{\mathrm{mot}} \mathbf{1}_{\mathbb{F}_p})/p), X \right).$$

The representing object is not compact by Theorem 6.1, so the functor does not preserve all filtered colimits. It is exact. If it preserved all coproducts, it would preserve all colimits, contradicting the first assertion. $\square$

Proof of Theorem 7.1. By Lemma 7.3, choose a small family $\{X_i\}_{i \in I}$ in $\mathrm{Mot}_{\mathbb{F}_p}^{\mathrm{loc}}$ on which the displayed functor fails to preserve the coproduct. By Lemma 7.2, choose small $\mathbb{F}_p$ -linear categories $\mathcal{C}_i$ with $U_{\mathrm{loc}}(\mathcal{C}_i) \simeq X_i$, and form

$$\mathcal{C} = \bigoplus_{i \in I} \mathcal{C}_i.$$

This is a small $\mathbb{F}_p$ -linear stable category. The functor U_{loc} preserves finite sums and filtered colimits, hence this coproduct. Thus $U_{\mathrm{loc}}(\mathcal{C}) \simeq \bigoplus_i X_i$. Using Proposition 5.1, the chosen failure of coproduct preservation becomes the statement that

$$(7.3) \quad \bigoplus_{i \in I} K_{\mathrm{mot}}^{\mathrm{Sel}}(\mathcal{C}_i)/p \longrightarrow K_{\mathrm{mot}}^{\mathrm{Sel}}(\mathcal{C})/p$$

is not an equivalence.

Suppose that $\alpha_{\mathcal{C}}/p$ were an equivalence. The inclusion of $\mathcal{C}_i$ into $\mathcal{C}$ and the projection onto its i -th summand exhibit $\mathcal{C}_i$ as a retract of $\mathcal{C}$. Naturality makes $\alpha_{\mathcal{C}_i}/p$ a retract of $\alpha_{\mathcal{C}}/p$, so every $\alpha_{\mathcal{C}_i}/p$ would also be an equivalence. The vertical maps in the natural square

$$\begin{array}{ccc} \bigoplus_i K_{\mathrm{mot}}^{\mathrm{Sel}}(\mathcal{C}_i)/p & \longrightarrow & K_{\mathrm{mot}}^{\mathrm{Sel}}(\mathcal{C})/p \\ \bigoplus_i (\alpha_{\mathcal{C}_i}/p) \downarrow & & \downarrow \alpha_{\mathcal{C}}/p \\ \bigoplus_i K_{\mathrm{mot}}^{\mathrm{Sel}}(\mathcal{C}_i)/p & \longrightarrow & K_{\mathrm{mot}}^{\mathrm{Sel}}(\mathcal{C})/p \end{array}$$

would therefore be equivalences. The bottom map is an equivalence by Proposition 5.1 and Corollary 5.3. This forces (7.3) to be an equivalence, a contradiction. The same map fails after derived p -completion, since equivalences after completion are detected modulo p .

The family $\{\mathcal{C}_i\}$ also shows that $K_{\mathrm{mot}}^{\mathrm{Sel}}(-)/p$ does not preserve all coproducts. The identification $K^{\mathrm{Sel}}(-)/p \simeq \mathrm{TC}(-)/p$ shows that $K^{\mathrm{Sel}}(-)/p$ preserves them. Since coproduct

preservation is invariant under natural equivalence, the two functors are not naturally equivalent. $\square$

Corollary 7.4 (A one-algebra counterexample). *For every prime p , there is a nonconnective E_1 - $H\mathbb{F}_p$ -algebra B such that $\alpha_{\mathrm{Perf}(B)}/p$ is not an equivalence.*

Proof. Apply Proposition A.1 in Appendix A to the category of Theorem 7.1. The resulting localization sequence has an exact section, and the induced splitting identifies the comparison at $\mathrm{Perf}(B)$ with the direct sum of the comparisons at $\mathcal{C}$ and $\mathrm{Perf}(\mathbb{F}_p)$. The latter is an equivalence by Corollary 3.8, whereas the former fails modulo p . Thus the comparison at B fails modulo p . By the same corollary, B cannot be connective. $\square$

Remark 7.5. Every $\mathbb{F}_p$ -linear category is $\mathbb{Z}$ -linear. Thus Theorem 7.1 also rules out a universal comparison on $\mathbb{Z}$ -linear categories. The obstruction is the continuity discrepancy in characteristic p .

8. CHROMATIC LOCALIZATION AND A PERIODIC EXAMPLE

Fix a prime p . For $n \geq 1$, let $T(n)$ be the telescope of a v_n -self-map of a finite type- n spectrum, considered up to Bousfield equivalence.

Theorem 8.1 (Higher-height comparison). *For every small stable category $\mathcal{C}$ and every $n \geq 2$, the maps $K(\mathcal{C}) \rightarrow K_{\mathrm{mot}}^{\mathrm{Sel}}(\mathcal{C})$ and $K^{\mathrm{Sel}}(\mathcal{C}) \rightarrow \mathrm{TC}(\mathcal{C})$ induce natural equivalences*

$$\begin{aligned} L_{T(n)} K_{\mathrm{mot}}^{\mathrm{Sel}}(\mathcal{C}) &\simeq L_{T(n)} K(\mathcal{C}), \\ L_{T(n)} K^{\mathrm{Sel}}(\mathcal{C}) &\simeq L_{T(n)} \mathrm{TC}(\mathcal{C}). \end{aligned}$$

Under these equivalences, $L_{T(n)} \alpha_{\mathcal{C}}$ is the localized cyclotomic trace.

Lemma 8.2. *For $n \geq 2$, we have*

$$L_{T(n)} \mathrm{End}_{\mathrm{Mot}^{\mathrm{loc}}}(L_{\mathbb{A}^1}^{\mathrm{mot}} \mathbf{1}) = 0.$$

Proof. The localization adjunction gives

$$\begin{aligned} \mathrm{End}_{\mathrm{Mot}^{\mathrm{loc}}}(L_{\mathbb{A}^1}^{\mathrm{mot}} \mathbf{1}) &\simeq \mathrm{map}_{\mathrm{Mot}^{\mathrm{loc}}}(\mathbf{1}, L_{\mathbb{A}^1}^{\mathrm{mot}} \mathbf{1}) \\ &\simeq |K(\mathbb{S}[\Delta^\bullet])|. \end{aligned}$$

The second equivalence follows from Proposition 2.2(i) and the fact that mapping from the compact unit $\mathbf{1}$ commutes with colimits.

The realization $|\mathbb{S}[\Delta^\bullet]|$ in connective ring spectra is the zero ring spectrum. Indeed, its zeroth homotopy ring is the coequalizer of $\mathbb{Z}[x] \rightrightarrows \mathbb{Z}$, with x sent to 0 and 1. Hence its unit is zero in π_0 . The unit action then makes the identity of $|\mathbb{S}[\Delta^\bullet]|$ nullhomotopic, so $|\mathbb{S}[\Delta^\bullet]| = 0$.

For $n \geq 2$, the functor $L_{T(n)} K$ on ring spectra preserves sifted colimits, computed in $T(n)$ -local spectra [LMMT, Corollary 4.32]. Therefore

$$\begin{aligned} L_{T(n)} \mathrm{End}_{\mathrm{Mot}^{\mathrm{loc}}}(L_{\mathbb{A}^1}^{\mathrm{mot}} \mathbf{1}) &\simeq L_{T(n)} |K(\mathbb{S}[\Delta^\bullet])| \\ &\simeq L_{T(n)} K(|\mathbb{S}[\Delta^\bullet]|) = 0. \end{aligned}$$

$\square$

Proof of Theorem 8.1. For every unital right module V over $\mathrm{End}_{\mathrm{Mot}^{\mathrm{loc}}}(L_{\mathbb{A}^1}^{\mathrm{mot}} \mathbf{1})$, the unit and action give a retraction

$$V \longrightarrow V \otimes \mathrm{End}_{\mathrm{Mot}^{\mathrm{loc}}}(L_{\mathbb{A}^1}^{\mathrm{mot}} \mathbf{1}) \longrightarrow V.$$

After tensoring with $T(n)$, the middle term vanishes by Lemma 8.2, so $T(n) \otimes V = 0$. Moreover, $T(n) \otimes L_{KU} \mathbb{S} = 0$ for $n \geq 2$. Since L_{KU} is smashing, $L_{KU} V$ is also $T(n)$ -acyclic, and hence so is $\mathrm{fib}(V \rightarrow L_{KU} V)$.

Precomposition makes

$$\mathrm{map}_{\mathrm{Mot}^{\mathrm{loc}}}(L_{\mathbb{A}^1}^{\mathrm{mot}} \mathbf{1}, U_{\mathrm{loc}}(\mathcal{C}))$$

a right module over $\text{End}_{\text{Mot}^{\text{loc}}} (L_{\mathbb{A}^1}^{\text{mot}} \mathbf{1})$. The preceding argument and (3.7) therefore show that $K(\mathcal{C}) \rightarrow K_{\text{mot}}^{\text{Sel}}(\mathcal{C})$ is a $T(n)$ -equivalence.

Every L_{KU} -local spectrum is $T(n)$ -acyclic. Applying the exact functor $L_{T(n)}$ to the pullback (1.1) proves the second equivalence. The composite

$$K(\mathcal{C}) \longrightarrow K_{\text{mot}}^{\text{Sel}}(\mathcal{C}) \xrightarrow{\alpha_{\mathcal{C}}} K^{\text{Sel}}(\mathcal{C}) \longrightarrow \text{TC}(\mathcal{C})$$

is the cyclotomic trace by the construction of $\alpha_{\mathcal{C}}$. This identifies the localized comparison map. $\square$

Lemma 8.3. *Let L be a smashing localization of spectra. For every cyclotomic spectrum X whose underlying spectrum is L -local, the spectrum $\text{TC}(X)$ is L -local.*

Proof. Local spectra are closed under limits and, since L is smashing, under colimits computed in spectra. They are therefore closed under homotopy orbits, homotopy fixed points, and finite-group Tate constructions. By [NS, Proposition II.1.9], $\text{TC}(X)$ is the fiber of

$$X^{h\mathbb{T}} \longrightarrow \prod_{q \text{ prime}} (X^{tC_q})^{h(\mathbb{T}/C_q)},$$

where the map is the difference between the canonical map and the Frobenius maps. Both terms are L -local, so their fiber is local. $\square$

Corollary 8.4 (A periodic counterexample). *At every prime p , we have*

$$\begin{aligned} L_{T(2)} K_{\text{mot}}^{\text{Sel}}(\text{Perf}(KU_p^{\wedge})) &\simeq L_{T(2)} K(KU_p^{\wedge}) \neq 0, \\ L_{T(2)} K^{\text{Sel}}(\text{Perf}(KU_p^{\wedge})) &= 0. \end{aligned}$$

Consequently, $K_{\text{mot}}^{\text{Sel}}(\text{Perf}(KU_p^{\wedge}))$ and $K^{\text{Sel}}(\text{Perf}(KU_p^{\wedge}))$ are inequivalent, both before and after p -completion.

Proof. The commutative ring spectrum $KU_p^{\wedge}$ is L_{KU} -local. For every $KU_p^{\wedge}$ -module V , we have

$$(L_{KU} \mathbb{S}) \otimes V \simeq ((L_{KU} \mathbb{S}) \otimes KU_p^{\wedge}) \otimes_{KU_p^{\wedge}} V \simeq V.$$

In particular, the underlying spectrum of $\text{THH}(KU_p^{\wedge})$ is L_{KU} -local. Lemma 8.3 gives $\text{TC}(KU_p^{\wedge}) \simeq L_{KU} \text{TC}(KU_p^{\wedge})$, and (1.1) becomes

$$K^{\text{Sel}}(\text{Perf}(KU_p^{\wedge})) \simeq L_{KU} K(KU_p^{\wedge}).$$

Its $T(2)$ -localization is zero.

The height-one Lubin–Tate spectrum $KU_p^{\wedge}$ satisfies $L_{T(2)} K(KU_p^{\wedge}) \neq 0$ by [Yua, Theorem A]. Theorem 8.1 gives the asserted nonvanishing for $K_{\text{mot}}^{\text{Sel}}(\text{Perf}(KU_p^{\wedge}))$. Finally, the derived completion map $X \rightarrow X_p^{\wedge}$ is a $T(n)$ -equivalence for every $n \geq 1$ [LMMT, Lemma 2.2(vi)]. The same distinction therefore holds after p -completion. $\square$

Remark 8.5. The category $\text{Perf}(KU_p^{\wedge})$ is not $\mathbb{Z}$ -linear. Such a structure would make the endomorphism spectrum

$$\text{End}_{\text{Perf}(KU_p^{\wedge})}(KU_p^{\wedge}) \simeq KU_p^{\wedge}$$

an $H\mathbb{Z}$ -module. The bounded-above periodic-homotopy vanishing in [LMMT, Lemma 2.2] gives $K(1) \otimes H\mathbb{Z} = 0$, so every $H\mathbb{Z}$ -module is $K(1)$ -acyclic. This contradicts the fact that $KU_p^{\wedge}$ is a nonzero $K(1)$ -local spectrum.

APPENDIX A. FROM A SMALL LINEAR CATEGORY TO ONE ALGEBRA

Proposition A.1. *Let k be a commutative ring and let $\mathcal{C}$ be a small k -linear idempotent-complete stable category. There is an augmented E_1 -Hk-algebra $B \rightarrow Hk$, split by its unit, and a Karoubi localization sequence*

$$\mathcal{C} \longrightarrow \text{Perf}(B) \longrightarrow \text{Perf}(k).$$

The last functor has an exact k -linear section. In particular,

$$U_{\text{loc}}(\text{Perf}(B)) \simeq U_{\text{loc}}(\mathcal{C}) \oplus \mathbf{1}_k.$$

The same splitting holds, naturally, after applying any localizing invariant.

Proof. We construct B by adjoining a unit to the finite-support matrix algebra of an Hk -enriched model for $\mathcal{C}$. Its matrix idempotents generate the kernel of extension of scalars to Hk ; their mapping spectra then identify the compact objects of this kernel with $\mathcal{C}$. This gives the localization sequence. Extension of scalars along the unit $Hk \rightarrow B$ supplies its section and the splitting of invariants.

Choose a small Hk -enriched category $\mathcal{A}$ with $\text{Perf}(\mathcal{A}) \simeq \mathcal{C}$, using the Morita comparison in Section 2. Define the nonunital Hk -algebra

$$I = \bigoplus_{x,y \in \text{Ob}(\mathcal{A})} \mathcal{A}(x,y)$$

by composition when the matrix indices match and by zero otherwise. The identities of $\mathcal{A}$ determine mutually orthogonal idempotents $e_x \in \pi_0 I$, whose corner spectra satisfy $e_y I e_x \simeq \mathcal{A}(x,y)$. Form the unitization

$$B = Hk \oplus I.$$

Its multiplication is induced by multiplication on Hk , the two scalar actions on I , and multiplication on I . The projection $B \rightarrow Hk$ is an augmentation, split by the unit $Hk \rightarrow B$.

Each right module $e_x B$ is an idempotent summand of B , hence compact. The finite-support condition gives decompositions

$$I \simeq \bigoplus_x B e_x \quad \text{as a left module}, \quad I \simeq \bigoplus_x e_x B \quad \text{as a right module}.$$

Each $B e_x$ is an idempotent summand of the left regular module. Since e_x maps to zero under the augmentation,

$$Hk \otimes_B^{\mathbb{L}} B e_x = 0, \quad Hk \otimes_B^{\mathbb{L}} I \simeq \bigoplus_x (Hk \otimes_B^{\mathbb{L}} B e_x) = 0.$$

Tensoring the bimodule triangle $I \rightarrow B \rightarrow Hk$ on the left with Hk shows that the multiplication map is an equivalence:

$$(A.1) \quad Hk \otimes_B^{\mathbb{L}} Hk \simeq Hk.$$

For every right Hk -module V , the adjunction counit is therefore an equivalence

$$V \otimes_B^{\mathbb{L}} Hk \simeq V \otimes_{Hk}^{\mathbb{L}} (Hk \otimes_B^{\mathbb{L}} Hk) \simeq V.$$

Restriction of scalars $\text{Mod}_{Hk} \rightarrow \text{Mod}_B$ is thus fully faithful, so extension of scalars is a localization.

Its kernel is

$$\mathcal{N} = \text{Loc}_{\text{Mod}_B} \{e_x B\}_x.$$

Every generator $e_x B$ is annihilated by extension of scalars, which gives one inclusion. Conversely, if $M \otimes_B^{\mathbb{L}} Hk = 0$, the bimodule triangle gives $M \simeq M \otimes_B^{\mathbb{L}} I$. The colimit-preserving exact endofunctor $-\otimes_B^{\mathbb{L}} I$ sends the generator B of Mod_B to $I \in \mathcal{N}$. Its image is consequently contained in $\mathcal{N}$, so $M \in \mathcal{N}$. This identifies the kernel.

The enriched mapping spectra of its compact generators are

$$\text{map}_{\text{Mod}_B}(e_x B, e_y B) \simeq e_y B e_x \simeq \mathcal{A}(x,y),$$

with composition induced by that of $\mathcal{A}$. Enriched Yoneda and Morita theory therefore identify $\mathcal{N}^\omega \simeq \text{Perf}(\mathcal{A}) \simeq \mathcal{C}$. Since $\mathcal{N}$ is generated by compact objects of Mod_B , the localization theorem for compact objects identifies the idempotent completion of $\text{Perf}(B)/\mathcal{C}$ with $\text{Perf}(k)$; see [Nee, Theorem 2.1] and [BGT, Section 5.2, especially Proposition 5.15]. This is the required Karoubi localization sequence.

Finally, extension of scalars along $Hk \rightarrow B$ sends the generator Hk to B , so it preserves perfect objects. Its composite with extension of scalars along $B \rightarrow Hk$ is the identity. Thus it gives an exact k -linear section on perfect categories. After applying any localizing invariant, the localization sequence becomes a fiber sequence with a section, hence splits. Naturality of the splitting with respect to the invariant gives the last assertions. $\square$

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